

заг. 1.1)

2 ИИ  $\rightarrow$  1 зара  
3 Зел.  $\rightarrow$  2 зара  
4 сини  $\rightarrow$  3 зара

$$P(\text{зел} | \text{сума } 3) \stackrel{\text{Бейт}}{=} \frac{P(\text{сума } 3 | \text{зел}) P(\text{зел})}{P(\text{сума } 3)}$$

$$= \frac{\frac{2}{36} \cdot \frac{3}{9}}{\frac{1}{6} \cdot \frac{2}{9} + \frac{2}{36} \cdot \frac{3}{9} + \frac{1}{216} \cdot \frac{4}{9}}$$

$$= \frac{9}{28} \approx 32,14\%$$

$$\begin{aligned} 1.2) P(\max \geq 3) &= 1 - P(\text{всички са 1 или 2}) \\ &= 1 - \left(\frac{2}{6}\right)^5 \end{aligned}$$

За  $C := \{\text{сума число, което се дели на } 5\}$

$D := \{ \text{--- 11 --- на } 2 \}$

$$P(\bar{C}) = \left(\frac{5}{6}\right)^5; P(\bar{D}) = \left(\frac{1}{2}\right)^3; P(\bar{C} \cap \bar{D}) = \left(\frac{2}{6}\right)^5$$

и тогава

$$P(C \cap D) = P(C) + P(D) - P(C \cup D)$$

$$= 1 - P(\bar{C}) + 1 - P(\bar{D}) - (1 - P(\bar{C} \cap \bar{D}))$$

$$= 1 - \left(\frac{5}{6}\right)^5 - \left(\frac{1}{2}\right)^5 + \left(\frac{1}{6}\right)^5 \approx 56.70\%$$

$$3) P(X_3 = 2) = P(2 \text{ κυτταί } \text{ c } 3 \text{ τερνίκια u } 1 \text{ c } 1 \text{ τερνίκια})$$

$$= \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot \binom{4}{3} \cdot 5/2}{7^7} \approx 1.78\%$$

$$P(X_3 = 1) = P(3+4, 3+1+1+1+1, 3+2+2, 3+2+1+1)$$

$$= \frac{7 \cdot \binom{7}{3} \cdot 6}{7^2} + \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot 5 \cdot 4 \cdot 3}{7^7} + \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot \binom{4}{2} \cdot 5/2}{7^7}$$

$$+ \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot \binom{4}{2} \cdot 5 \cdot 4}{7^7} \approx 34.99\%$$

$$P(X_3 = 0) = 1 - P(X_3 = 0 \text{ u } 1) \approx 63.23\%$$

$$\Rightarrow E[X_3] = P(X_3 = 1) + 2P(X_3 = 2)$$

$$\approx 0.3855$$

$$\sum_{i=1}^7 i E[X_i] = E\left[\sum_{i=0}^7 X_i\right] = E[\text{σπου τερνίκια}]$$

$$= 7$$

Αθροίζουμε  $\sum_{i=0}^7 E[X_i] = E[X_0 + \dots + X_7] = E[\text{\# κυτταί}] = 7$ .

209.2  $Y := \# \text{πραβυλνι}$ , οτ τεζν, κατω ζκαε  $\sim \text{Bin}(10, \frac{9}{10})$

$Z :=$  ————— κατω τεζν καε  $\sim \text{Bin}(5, \frac{1}{5})$   
 $X = Y + Z$

$$2.1) P(X > 12) = \sum_{k=3}^5 P(Z=k) P(Y > 12-k)$$

$$= P(Z=3) [P(Y=10)]$$

$$+ P(Z=4) [P(Y=10) + P(Y=9)]$$

$$+ P(Z=5) [P(Y=10) + \dots + P(Y=8)]$$

$$E[X] = E[Y] + E[Z] = 10 \cdot \frac{9}{10} + 5 \cdot \frac{1}{5} = 10.$$

$$D[X] \stackrel{\text{leg.}}{=} D[Y] + D[Z] = 10 \cdot \frac{9}{10} \cdot \frac{1}{10} + 5 \cdot \frac{1}{5} \cdot \frac{4}{5} = \frac{17}{10}.$$

$$2.2) P(Y=3 | X=3) \stackrel{\text{Bayes}}{=} \frac{P(X=3 | Y=3) P(Y=3)}{\sum P(X=3 | Y=k) P(Y=k)}$$

$$= \frac{P(Z=0) P(Y=3)}{\sum P(Z=3-k) P(Y=k)}$$

$$= \frac{\left(\frac{4}{5}\right)^5 \cdot \binom{10}{3} \left(\frac{9}{10}\right)^3 \left(\frac{1}{10}\right)^7}{\sum_{k=0}^3 \binom{5}{3-k} \left(\frac{1}{5}\right)^{3-k} \left(\frac{4}{5}\right)^{k+2} \binom{10}{k} \left(\frac{9}{10}\right)^k \left(\frac{1}{10}\right)^{10-k}}$$

$$= \frac{\left(\frac{4}{5}\right)^5 \cdot \binom{10}{3} \left(\frac{9}{10}\right)^3 \left(\frac{1}{10}\right)^7}{\sum_{k=0}^3 \binom{5}{3-k} \left(\frac{1}{5}\right)^{3-k} \left(\frac{4}{5}\right)^{k+2} \binom{10}{k} \left(\frac{9}{10}\right)^k \left(\frac{1}{10}\right)^{10-k}} \dots$$

geg. 3

| $X \backslash Y$ | 0             | 1             | 2             |     |
|------------------|---------------|---------------|---------------|-----|
| 0                | $\frac{1}{9}$ | 0             | $\frac{2}{9}$ | 1/3 |
| 1                | $\frac{2}{9}$ | $\frac{1}{9}$ | 0             | 1/3 |
| 2                | 0             | $\frac{2}{9}$ | $\frac{1}{9}$ | 1/3 |
|                  | $\frac{1}{3}$ | $\frac{1}{3}$ | $\frac{1}{3}$ |     |

$$EX = EY = 1; E[XY] = 1 \cdot 1 \cdot \frac{1}{9} + 2 \cdot 1 \cdot \frac{2}{9} + 2 \cdot 2 \cdot \frac{1}{9} = 1$$

$$\Rightarrow \text{Cov}(X, Y) = E[XY] - E[X] \cdot E[Y] = 0$$

$$\Rightarrow \boxed{\text{Cov}(X, Y) = 0}$$

$$2. g_X(s) = \frac{1}{6}(s + \dots + s^6)$$

$$g'_X(s) = \frac{1}{6}(1 + 2s + \dots + 6s^5) \Rightarrow g'_X(1) = \frac{7}{2}$$

$$g''_X(s) = \frac{1}{6}(2 + 6s + 12s^2 + 20s^3 + 30s^4) \Rightarrow g''_X(1) = \frac{70}{6}$$

$$\Rightarrow E[X] = g'_X(1) = \frac{7}{2}$$

$$DX = g''_X(1) + g'_X(1) - g'_X(1)^2$$

$$= \frac{70}{6} + \frac{7}{2} - \frac{49}{4} = \frac{35}{12}$$

3. Also  $X \perp Y$ , so  $P(X=k, Y=m) = P(X=k)P(Y=m)$   
 u. genau

$$g_{X+Y}(s) = \sum_{k=0}^{\infty} s^k P(X+Y=k)$$

$$\stackrel{X \perp Y}{=} \sum_{k=0}^{\infty} s^k \sum_{m=0}^k P(X=m) P(Y=k-m)$$

$$= \sum_{m=0}^{\infty} \sum_{k=m}^{\infty} s^m P(X=m) s^{k-m} P(Y=k-m)$$

$$\stackrel{i=k-m}{=} \sum_{m=0}^{\infty} s^m P(X=m) \sum_{i=0}^{\infty} s^i P(Y=i)$$

$$= g_X(s) g_Y(s).$$

Обратното не е вярно:

$$\text{Ако } g_{X+Y}(s) = g_X(s) g_Y(s) \not\Rightarrow X \perp Y$$

Контрпример са  $X$  и  $Z$  от задача 1  
Проверете!

зг. 4 Нека  $p_n(k) := P(k \in E_n)$

$$= P(\text{избран сме } k \text{ / използвайки} \\ \text{с избор от } [1, n])$$

Ключово наблюдение е, че :

$$p_n(k) = \underbrace{p_n(k+1)}_{\substack{\text{uzdaryame} \\ k, \text{ del'ga} \\ \text{uzdaryame } k+1 \\ (\text{zaygo tazu} \\ \text{b-it e } p_n(k+1)?)}} + \underbrace{p_n(k+1)}_{\substack{\text{uzdaryame} \\ k+1}} \cdot \underbrace{\frac{1}{k}}_{\substack{\text{na} \\ \text{cheg} \\ \text{ob } k}}$$

$$\Rightarrow p_n(k) = \frac{k+1}{k} p_n(k+1)$$

Тогда как  $p_n(n) = \frac{1}{n}$

$$\Rightarrow p_n(k) = \frac{k+1}{k} p_n(k+1) = \frac{k+1}{k} \cdot \frac{k+2}{k+1} p_n(k+2)$$

$$= \dots = \frac{\cancel{k+1}}{k} \cdot \frac{\cancel{k+2}}{\cancel{k+1}} \cdot \dots \cdot \frac{\cancel{n}}{\cancel{n-1}} p_n(n)$$

$$= \boxed{\frac{1}{k}}$$

2)  $P(2 \in E_n | 3 \notin E_n) = \frac{p_n(3)}{1 - p_n(3)} = \frac{1/3}{2/3} = \frac{1}{2}$  ↗ тут по-моему загвоздка

3) Here  $X_i = 1_{\{\text{uzdaryame me zuchu iz}\}} \sim \text{Ber}(\frac{1}{i})$

$$\Rightarrow E[|E_n|] = E[X_1 + \dots + X_n]$$

$$\stackrel{\text{линейность}}{=} \sum E[X_i]$$

$$= 1 + \frac{1}{2} + \dots + \frac{1}{n}.$$

Бележка 1: Оказва се, че  $X_i \perp X_j$  за  $i \neq j$ !

Бел. 2: Връзката тази процедура е обратната на Reservoir Sampling (виз. упрямкени), което е обратното отговора и Бел. 1.